

1. Löse jeweils nach allen enthaltenen Symbolen auf.

a) $x = \frac{y}{z}$

Lösung:

$$y = xz; z = \frac{y}{x}$$

b) $\varrho = \frac{m}{V}$

Lösung:

$$m = \varrho V; V = \frac{m}{\varrho}$$

c) $A = BC$

Lösung:

$$B = \frac{A}{C}; C = \frac{A}{B}$$

d) $a = \frac{b}{c} + d$

Lösung:

$$d = a - \frac{b}{c}; b = (a-d)c;$$

$$c = \frac{b}{a-d}$$

e) $a(b+c) = d$

Lösung:

$$a = \frac{d}{b+c}; b = \frac{d}{a} - c;$$

$$c = \frac{d}{a} - b$$

f) $ab + c = d$

Lösung:

$$a = \frac{d-c}{b}; b = \frac{d-c}{a}; c = d - ab$$

g) $x^2 = y$

Lösung:

$$x = \sqrt{y}$$

h) $x^2 + 1 = z$

Lösung:

$$x = \sqrt{z-1}$$

i) $\frac{1}{x} = y$

Lösung:

$$y = \frac{1}{x}$$

j) $E = \frac{1}{2}mv^2$

Lösung:

$$m = \frac{2E}{v^2}; v = \sqrt{\frac{2E}{m}}$$

k) $F = G \frac{m_1 m_2}{r^2}$

Lösung:

$$G = \frac{Fr^2}{m_1 m_2}; m_1 = \frac{Fr^2}{Gm_2}; m_2 = \frac{Fr^2}{Gm_1}; r = \sqrt{\frac{Gm_1 m_2}{F}}$$

l) $\frac{1}{x} = \frac{1}{y+z}$

Lösung:

$$x = y + z; y = x - z;$$

$$z = x - y$$

m) $\frac{1}{x} = \frac{1}{y} + \frac{1}{z}$

Lösung:

$$x = \frac{1}{\frac{1}{y} + \frac{1}{z}} = \frac{yz}{y+z};$$

$$y = \frac{1}{\frac{1}{x} - \frac{1}{z}} = \frac{xz}{z-x}; z = \frac{1}{\frac{1}{x} - \frac{1}{y}} = \frac{xy}{y-x}$$

n) $\frac{a+b}{c} = \frac{1}{d+e}$

Lösung:

$$a = \frac{c}{d+e} - b; b = \frac{c}{d+e} - a;$$

$$c = (a+b)(d+e); d = \frac{c}{a+b} - e; e = \frac{c}{a+b} - d$$

o) $T = 2\pi\sqrt{\frac{l}{g}}$

Lösung:

$$l = g \left(\frac{T}{2\pi}\right)^2; g = l \left(\frac{2\pi}{T}\right)^2$$